Open Source Hyperledger Projects and How They Can be Implemented

Hyperledger is an open source blockchain platform under the umbrella of the Linux Foundation. It contains multiple frameworks like Indy, Grid, Fabric, Iroha, Sawtooth and Burrow, with support tools like Composer, Aries, Caliper, Quilt, Transact, URSA, Explorer and Cello. Hyperledger's many frameworks are appropriate for various blockchain architectural purposes unlike other such platforms that do not have variant services.

Hyperledger has a unique implementation of functions called Chaincode, which are small pieces of code written in Java, NodeJS or GoLang (like a function in Azure or Lambda in AWS). These can be used to interact with the transactions handled in the smart ledger in a Hyperledger platform. By using this, we can define business logic, such as the execution of rules and the processing logic. Chaincode is therefore a very powerful component of the Hyperledger architecture.

Hyperledger architecture

Hyperledger architecture is unique in blockchain platforms with three-layered components — the infrastructure layer (which contains the ecosystem platform itself), the framework layer that contains the business blockchain frameworks for community development, and the tools layer that contains the accelerators suitable for these framework layers. The services integrated together in the Hyperledger architecture are shown in Figure 1.

The design philosophy of the Hyperledger frameworks is integrated into the various layers of services as listed below.

Consensus layer: This layer is the brain of the Hyperledger platform, which handles the business logic for transaction management. It ensures the transactions are executed correctly and that the blocks are handled properly.